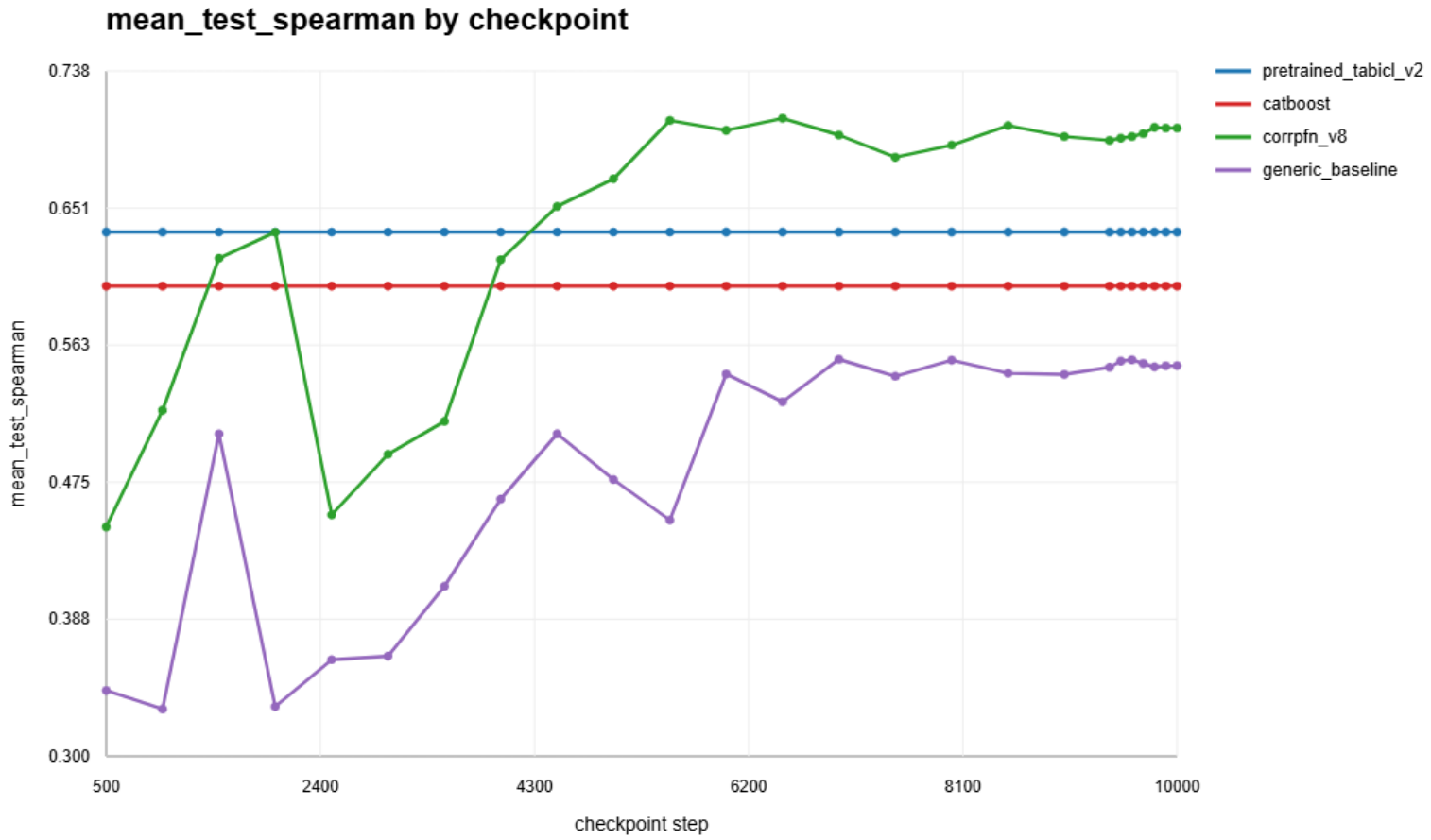
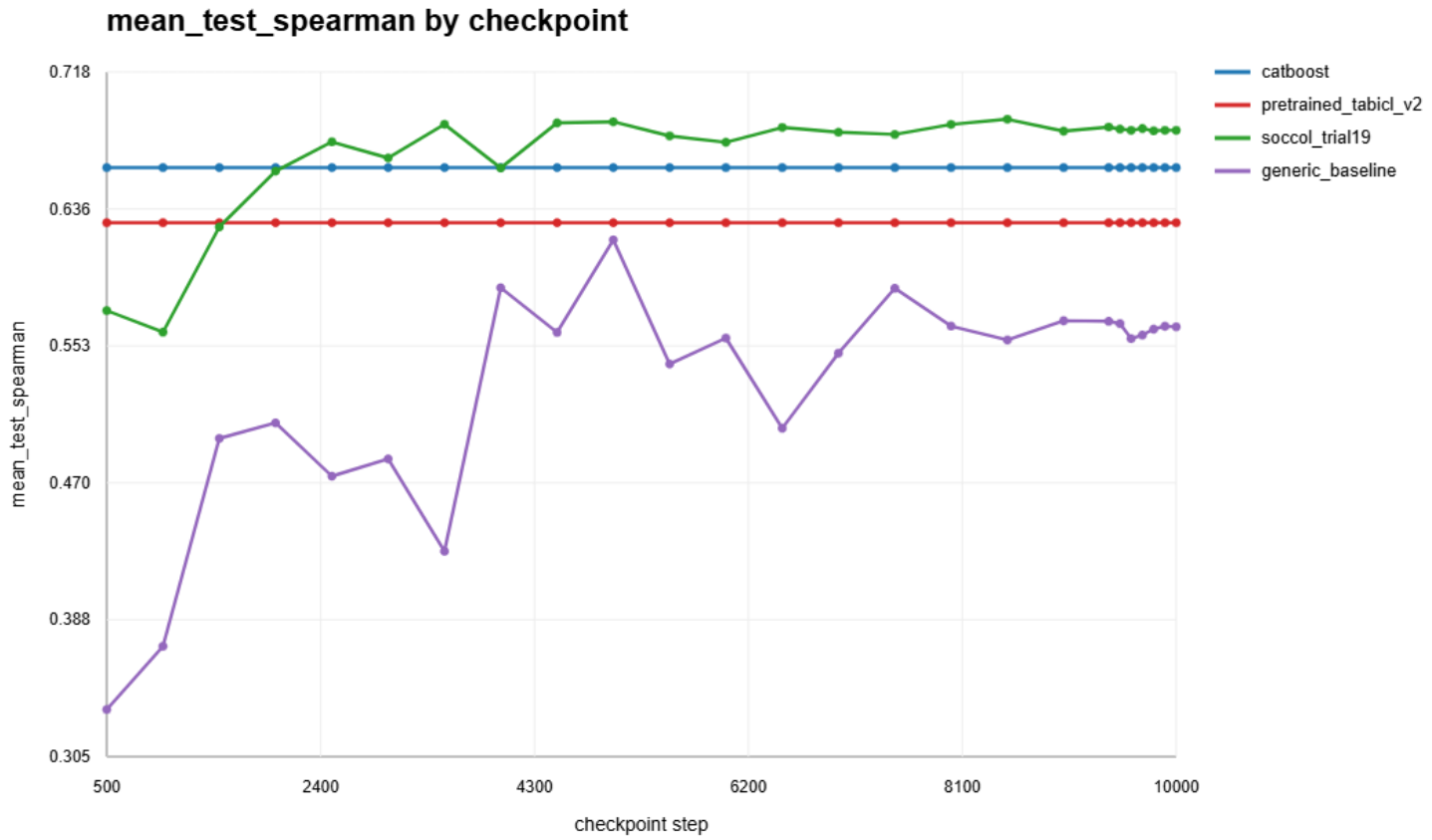
RECAP

- **EPIT folder:** [v8_final_test_cpu_comparison/](/home/fcolanto/projects/tabicl/corrosion_datasets/analysis/epit_pipeline/v8_final_test_cpu_comparison/)

- **Training EPIT and Soccol:** separate models trained for **10,000 steps** using configurations selected by Optuna on development folds.
- **EPIT configuration — Trial 17:** 50% generic + 50% informed; informed MLP/tree split **70/30**; formula weight **0.264**; composition perturbation **0.150**.
- **EPIT evaluation: 28 features**; 608 EPIT development rows as context → **152 fixed EPIT final-test queries**.



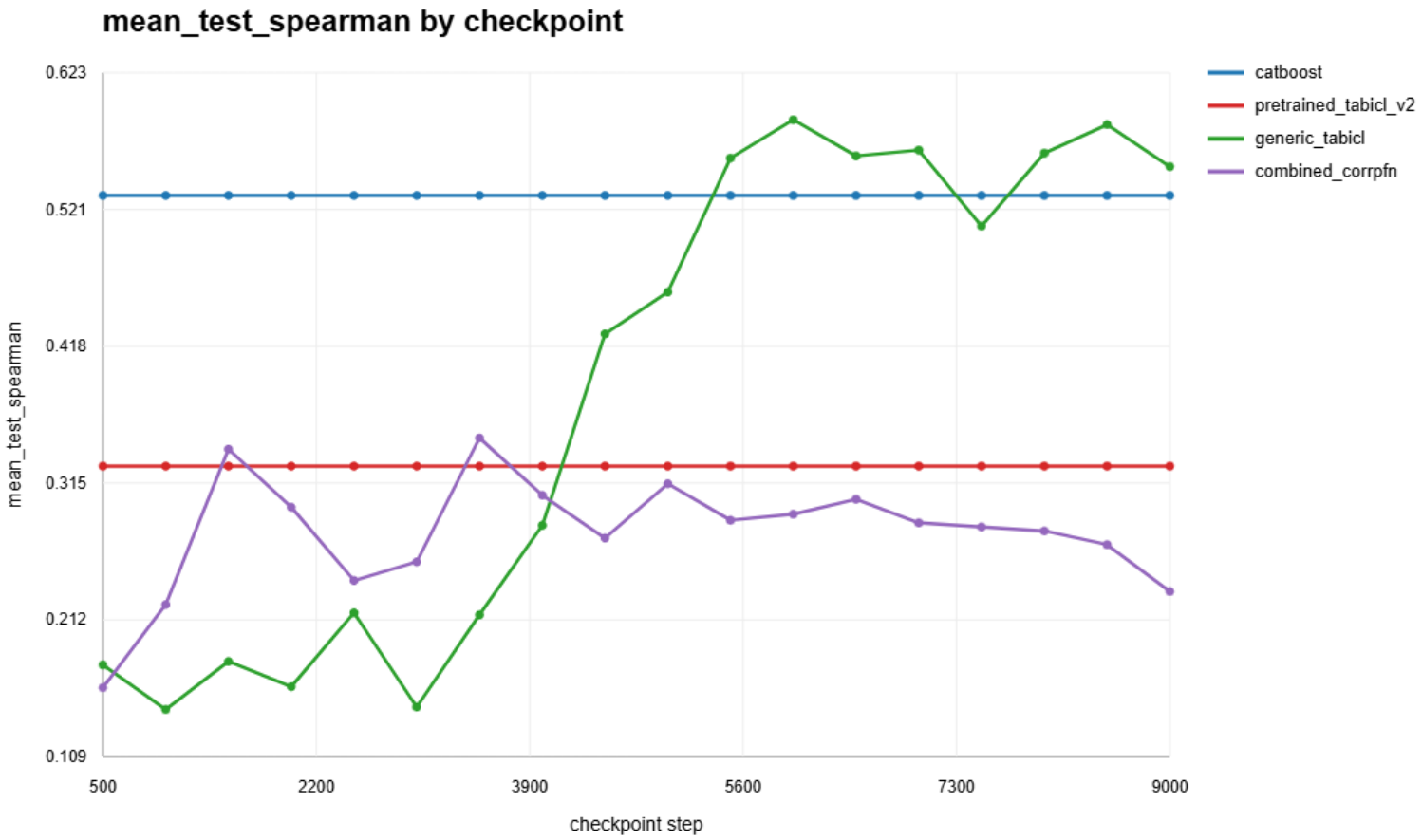
- **Soccol folder:** [final_test_full_development_comparison/](/home/fcolanto/projects/tabicl/corrosion_datasets/analysis/soccol_pipeline/final_test_full_development_comparison/)
- **Soccol configuration — Trial 19:** 100% informed; informed MLP/tree split **100/0**; formula weight **0.663**; composition perturbation **0.036**.
- **Soccol evaluation: 37 features**; 3,222 Soccol development rows as context → **805 fixed Soccol final-test queries**.
- **Both EPIT and Soccol:** coefficient variation **0.8**; compared against generic TabICL, pretrained TabICL, and CatBoost.



COMB Model

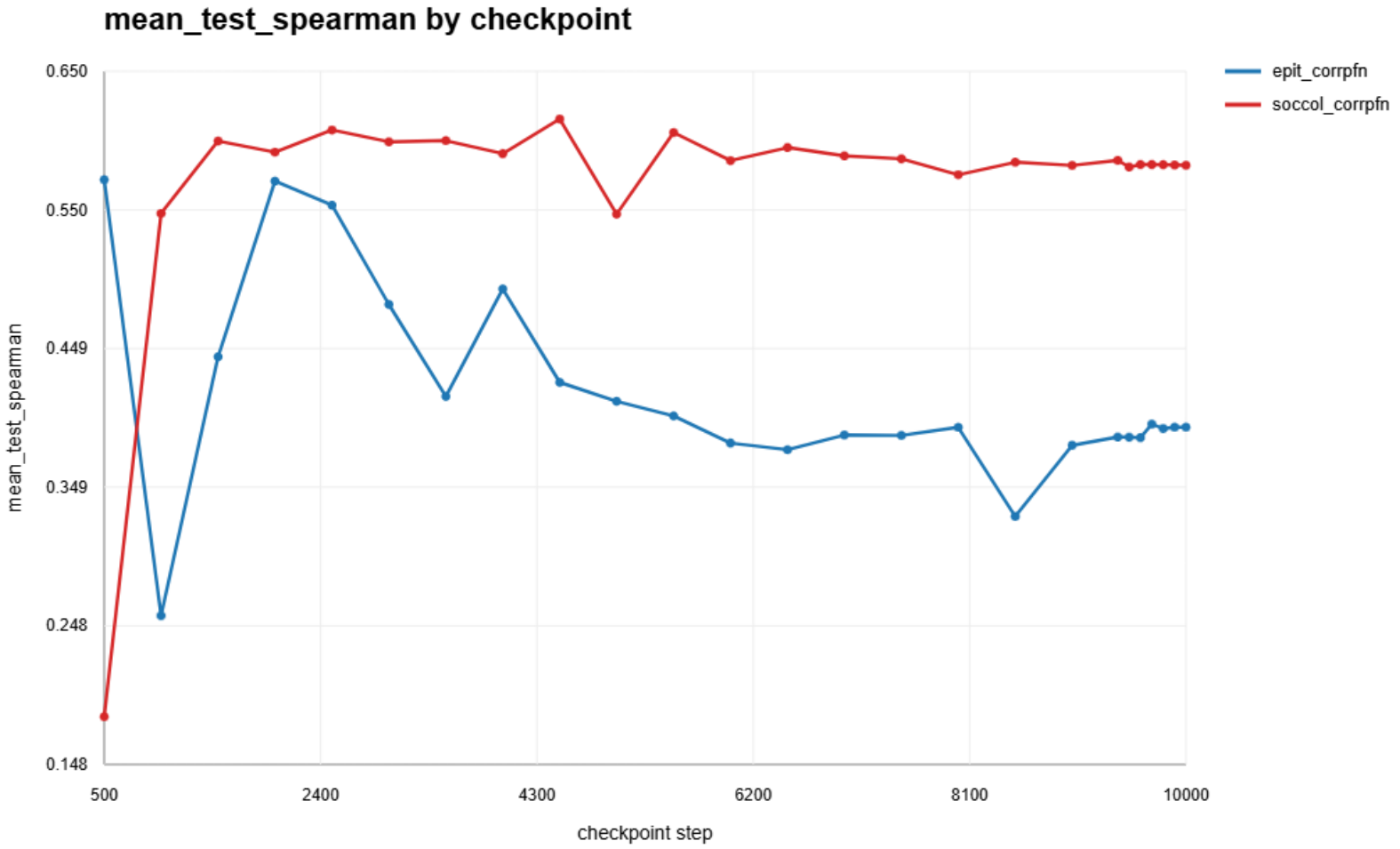
- **Folder:** [combined_v2_comparison_cpu/](/home/fcolanto/projects/tabicl/corrosion_datasets/analysis/combined_epit_soccol_mpea/combined_v2_comparison_cpu/)

- **Combined recipes equally:** EPIT (50% generic, 50% informed) + Soccol (100% informed).
- **One model; one prior per synthetic table:** 25% generic, 25% EPIT-informed, 50% Soccol-informed; original formulas/settings retained.
- **Inputs:** 50 shared feature columns; learned missingness vector.
- **Evaluation:** checkpoints through **9,000 steps**; 4,787 EPIT + Soccol context rows → 267 fixed MPEA queries.

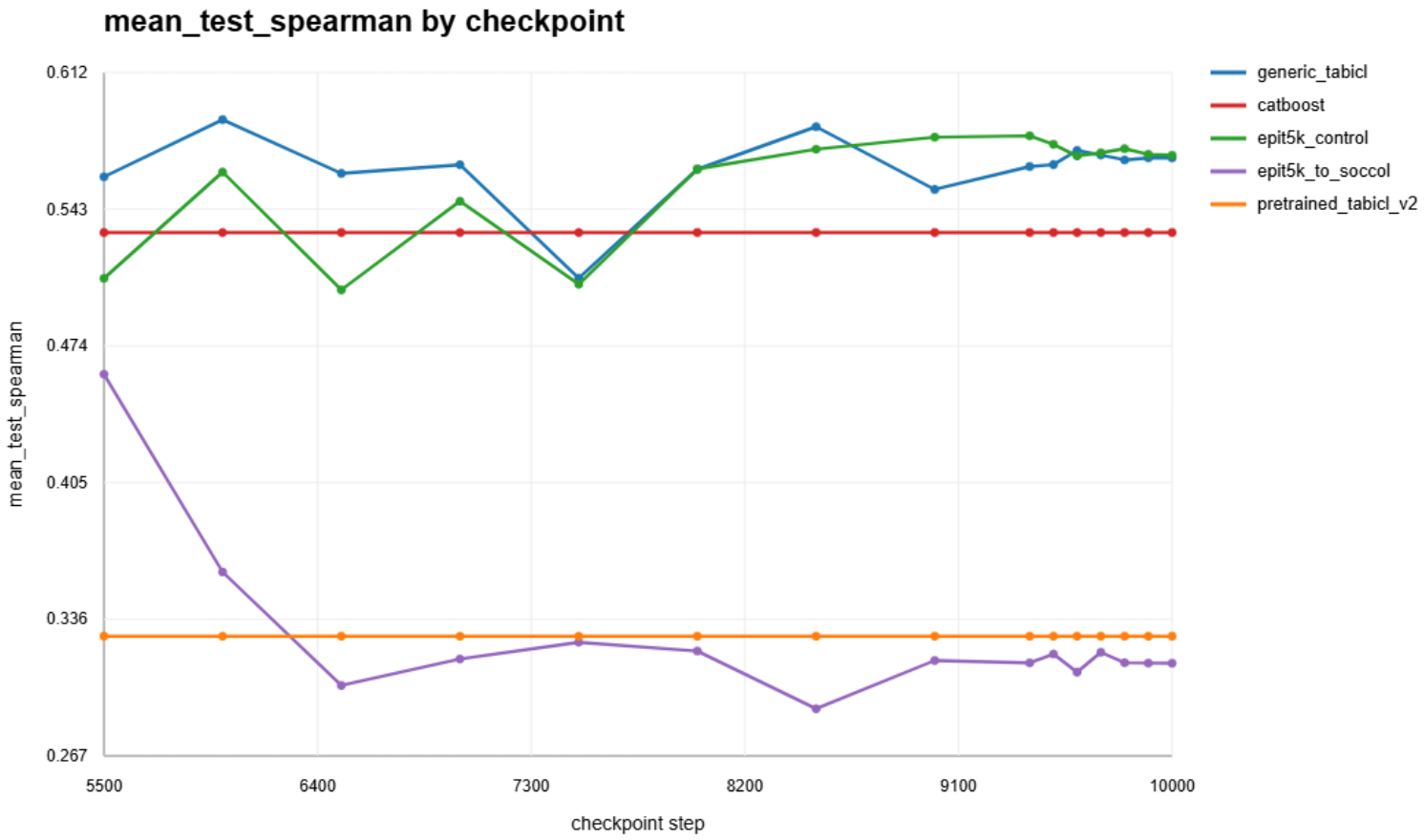


- **Folder:** [shared_specialist_comparison_cpu/](/home/fcolanto/projects/tabicl/corrosion_datasets/analysis/combined_epit_soccol_mpea/shared_specialist_comparison_cpu/)

- **Training:** reused combined's training code and architecture to train two separate models reproducing the specialist recipes.
- **Ratios:** EPIT model = 50% generic + 50% EPIT-informed; Soccol model = 100% Soccol-informed.
- **Inputs:** same 50 shared columns and missingness handling as combined.
- **Purpose:** test whether the specialist recipes work within the combined model's setup.



- **Folder:** [epit5k_continuation_comparison_cpu/](/home/fcolanto/projects/tabicl/corrosion_datasets/analysis/combined_epit_soccol_mpea/epit5k_continuation_comparison_cpu/)
- **Starting point:** shared-schema EPIT model at **5,000 steps**.
- **Two continuations:** keep **50% generic + 50% EPIT**, or switch to **100% Soccol**.
- **Same setup:** architecture, schema, missingness handling, and learning-rate schedule.
- **Purpose:** test whether switching to Soccol improves or damages the EPIT model's existing MPEA performance.



- **Folder:** [g50_epit25_soccol25_comparison_cpu/](/home/fcolanto/projects/tabicl/corrosion_datasets/analysis/combined_epit_soccol_mpea/g50_epit25_soccol25_comparison_cpu/)
- **New model trained from scratch: 50% generic, 25% EPIT-informed, 25% Soccol-informed**.
- **Change:** increased generic from 25% to 50%; reduced Soccol from 50% to 25%.
- **Unchanged:** source generators, architecture, shared schema, and missingness handling.
- **Purpose:** test whether more generic and less Soccol training improves MPEA performance.

